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$z = e^{it} \quad (0 \leq t < 2\pi)$ とおくと, $dz/dt = ie^{it}$ なので, $dz = izdt$.

$$
\begin{aligned}
\oint z^k dz &= i \oint z^{k+1} dt \\
&= i \oint e^{(k+1)it} dt \\
&= \frac{i}{(k+1)i} \left[e^{(k+1)it} \right]_0^{2\pi} \\
&= 0
\end{aligned}
$$

$$
\begin{aligned}
\oint \frac{1}{z} dz &= i \oint dt \\
&= i \left[t \right]_0^{2\pi} \\
&= 2\pi i
\end{aligned}
$$